

KARTHIK SRINIVASAN

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EDUCATION

Carnegie Mellon University (GPA 4.08/4.0)

Pittsburgh, PA

Master of Science, Robotic Systems Development

*May 2027**

Relevant Coursework: Generative AI, Robot Autonomy, Computer Vision, Robot Mobility, Manipulation and Control

Teaching assistantships: Fundamentals of Control, Fundamentals of Robot Control, Robot Kinematics and Dynamics*

Indian Institute of Technology Madras (IIT Madras) (GPA 9.36/10)

Chennai, India

Bachelor of Technology, Mechanical Engineering + Master of Technology, Data Science

Jul 2023

Relevant Coursework: Deep Learning, Modern Control Theory, Nonlinear Systems Analysis, Introduction to Robotics, Linear Algebra, Probability, Statistics and Stochastic Process

PUBLICATIONS

- Kharbanda, A., Wu, X., Muerto, J., **Srinivasan, K.**, Kangas, J. D., & Xu, Min (2026). *Orchestrating Classical Robotic Modules with Vision-Language Models: A Hybrid Architecture for Autonomous Laboratory*. **In submission to CoRL 2026.**
- **S. Karthik**, G. Rohith, K.B. Devika, S.C. Subramanian (2024), *A deep learning based encoder-decoder model for speed planning of autonomous electric truck platoons*, Elsevier Heliyon 10 e31836

SKILLSETS

Languages: Python (PyTorch, Keras, TensorFlow), C++; **Robotics:** MATLAB, ROS2, MoveIt, Gazebo, MuJoCo, Isaac Sim

INTERNSHIPS

MACHINE LEARNING INTERN, AUTONOMY | Serve Robotics, San Carlos, CA

May 2026 - Present

- **Developing a predictive pedestrian behavior model** using NVIDIA's Trace diffusion framework to forecast complex trajectories and dynamic obstacle interactions on sidewalks. **Training the diffusion model on 40+ hours of real-world deployment data** to capture realistic pedestrian movement patterns.
- Using the above model to **train an end-to-end Reinforcement Learning (RL) navigation policy** in simulation using a custom reward model, enabling the robot to perform socially aware maneuvers like yielding on narrow sidewalks.

DEEP LEARNING INTERN | Mathlogic Consulting, Gurugram, India

May 2022 - Jul 2022

- Performed quantitative analysis (Normalization, PCA, correlation) on an imbalanced dataset, reducing the features by **15%**.
- Trained a **Transformer with Self Attention and Inter-Sample Attention** for insurance claims approval prediction (imbalanced dataset). Attained **7.5% and 10% improvement** in ROC-AUC and F1 score over incumbent XGBoost models.

DAAD-WISE RESEARCH INTERN | Control Systems Group, TU Berlin, Germany

Jun 2021 - Aug 2021

- Developed a multi-agent **Iterative Learning Controller** where robots learned from each other to perform a limbo.
- **Achieved 35% performance gain** by weighting each robot's contribution by inverse error for every learning iteration.

PROJECTS

PHYSICS-AWARE REWARD GUIDANCE FOR TEXT-TO-MOTION GENERATION | GenAI Course, CMU

Mar 2026 - May 2026

- Infused a **reward-model framework** into a human **motion diffusion model** to improve physical plausibility of trajectories.
- Trained a time-conditioned Transformer with AdaLN via a Bradley-Terry objective on a generated contrastive dataset of valid and invalid motions. **Reduced the floating metric by 26%**, placing **3rd among 20 teams** in the course poster session.

SEMANTIC SEGREGATION USING VLMs FOR A FRANKA ARM | Robot Autonomy Course, CMU

Mar 2026 - May 2026

- Developed a **semantic segregation system using VLMs** to determine appropriate rack placement for household items, with a motion planner executing trajectories on a Franka Arm.
- Benchmarked **Qwen3-VL, Claude 3.5 Sonnet, and Gemini 2.5 Pro** across 10 objects, achieving **90% accuracy** with Gemini.
- Implemented **PRM motion planner with trajectory caching**, validating in **MuJoCo** before real-robot deployment.

AUTONOMOUS LABORATORY ASSISTANT ROBOT | Capstone Project, CMU (Guide: Dr. Min Xu)

Aug 2025 - Present

- Manufacturing the mobile base for mounting an **xArm6 robot** using swerve drives and Aluminum tubes in-house.
- Implementing the **autonomy stack** for a robotic system to **autonomously conduct experiments** using an **xARM6 robot**.
- **Developing** perception pipeline (**YOLO26**) for **real-time 6-DOF pose estimation** of liquid handlers and assay plates.
- Implementing **motion planning (RRT*, CHOMP)**, using **ROS2-MoveIt** for collision and **spillage-free trajectory** generation.
- Among top 20 teams to obtain the **\$1000 Gebhardt Sandbox Fund** (Swartz Entrepreneurship cell) for early-stage founders.

AI-DRIVEN TRUCK PLATOON CONTROLLER FRAMEWORK | Dual Degree Thesis, IIT Madras

May 2022 - May 2023

- Developed a **truck platoon framework** with an encoder-decoder **LSTM** to predict leader speed from battery, road, vehicle and charging stop constraints. Used train dataset of 2.1M points and obtained **3% MAPE** with test data (1.5M points).
- Implemented **string-stable Sliding Mode Controller** for follower vehicle tracking; Generated collision free velocity profiles.

CHARACTER RECOGNITION & OBJECT TRACKING | Computer Vision Course, CMU

Oct 2025 - Nov 2025

- Trained a CNN from scratch using **backprop** and **Xavier initialization**, achieving **75% accuracy** on the MNIST36 dataset.
- Developed an **end-to-end OCR pipeline** using **image morphology** to segment handwritten paragraphs with **70% accuracy**.
- Integrated a **FasterRCNN detector** with a custom **IoU tracker** to maintain consistent object IDs across video sequences.

3D SCENE RECONSTRUCTION & EPIPOLAR GEOMETRY | Computer Vision Course, CMU

Sep 2025 - Sep 2025

- Built a 3D reconstruction pipeline using the **Eight-Point Algorithm** and SVD to model a **temple** from image pairs.
- Recovered 6-DOF camera poses and 3D point clouds via **Essential Matrix** decomposition and triangulation.
- Developed a **RANSAC-based** estimator to reconstruct 3D keypoints for a **three-camera temporal sequence**.

WORK EXPERIENCE

HSBC Global Markets, Bengaluru, India

Jul 2023 - Jul 2025

Senior Associate, Equities Execution Services

- **Tasked with** improving execution performance. **Identified** hidden liquidity through market data research, heuristically **quantified** and **developed** a regression model to **capture 25% additional volume** without any negative market impact.
- **Explored** incorporating order aggression metrics into the market impact model through **weighted regression analysis**.

Associate, Equities Execution Services

- Led quantitative research on an emerging market with **non-continuous market data feed** and built a **regression model** using available parameters as input to estimate trade counts, enabling electronic execution in that market.
- Built a tool for **post trade analysis of orders** and to flag non-optimally executed orders to enable better troubleshooting.
- Awarded **Rising Star** of Q2 (2024) in the Markets and Securities Services division for exemplary performance.

LEADERSHIP

FORMULA SAE (RAFTAR FORMULA RACING), IIT Madras, Chennai

Team Captain

Apr 2021 - Mar 2022

- **Responsible** for recruitment, budget planning, project management and the entire functioning of the organization.
- **Led** 40+ members transitioning from gasoline to electric, **designed** team's 1st electric racecar with a **\$40K budget**.
- **Secured Overall championship** in the team's 1st competition as an electric team (Formula Bharat 21-22 - Virtuals).
- **Planned** expenditure for the year and **procured** critical components including motor, motor controller and charger.
- Formula Bharat 2021-22 (electric, virtual) - Led the team to **Championship victory**, the first competition with a fully-electric car design.